



Course: Applied Machine Learning for Business Analytics with Python

Course Code: BAN744

Project Title: Online Sales Data Analysis to Predict Revenue and Sales

Problem Type: Regression (Numerical Prediction)

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Methodology Summary

A. Data Pre-processing

- Date conversion to datetime format.
- IQR outlier detection on Total Revenue; cleaned dataset (df_cleaned) used for modelling.
- StandardScaler normalisation of numerical features for clustering and ML models.
- One-hot encoding of categorical features (Product Category, Region, Payment Method).

B. Machine Learning Models

- Linear Regression — baseline OLS model.
- Decision Tree Regressor — non-linear baseline.
- Random Forest Regressor — ensemble of 100 decision trees.
- Tuned Random Forest — GridSearchCV hyperparameter optimisation.
- Gradient Boosting Regressor — sequential ensemble; best overall R^2 .

C. Evaluation Metrics

- R^2 (coefficient of determination) — proportion of variance explained.
- MAE (Mean Absolute Error) — average absolute prediction error in USD.
- RMSE (Root Mean Squared Error) — penalises large prediction errors.
- MAPE (Mean Absolute Percentage Error) — scale-independent error measure.

D. Clustering

- Algorithm: K-Means (k=3), selected via elbow method / silhouette analysis.
- Features: Units Sold, Unit Price, Total Revenue + one-hot Product Category.
- Preprocessing: StandardScaler applied before clustering.

E. Dataset source: <https://www.kaggle.com/>

1. Executive Summary

This report presents a comprehensive analysis of online sales data spanning January through August 2024, covering 240 transactions across three global regions and six product categories. The analysis encompasses revenue performance, outlier investigation, customer segmentation, predictive modelling and actionable strategic recommendations.

Key findings from this analysis:

- Electronics dominates revenue, contributing 43.4% of total sales (\$34,982).
- North America leads regionally at \$36,844 (45.7% of global revenue).
- 19 high-value outlier transactions account for 39.7% of total revenue, concentrated in Electronics.
- Gradient Boosting and tuned Random Forest models achieved $R^2 > 0.994$, enabling highly reliable revenue forecasting.
- K-Means clustering identified three distinct customer segments for targeted marketing strategies.

2. Data Overview

The dataset comprises 240 transactions recorded between 1 January and 9 August 2024, with the following structure:

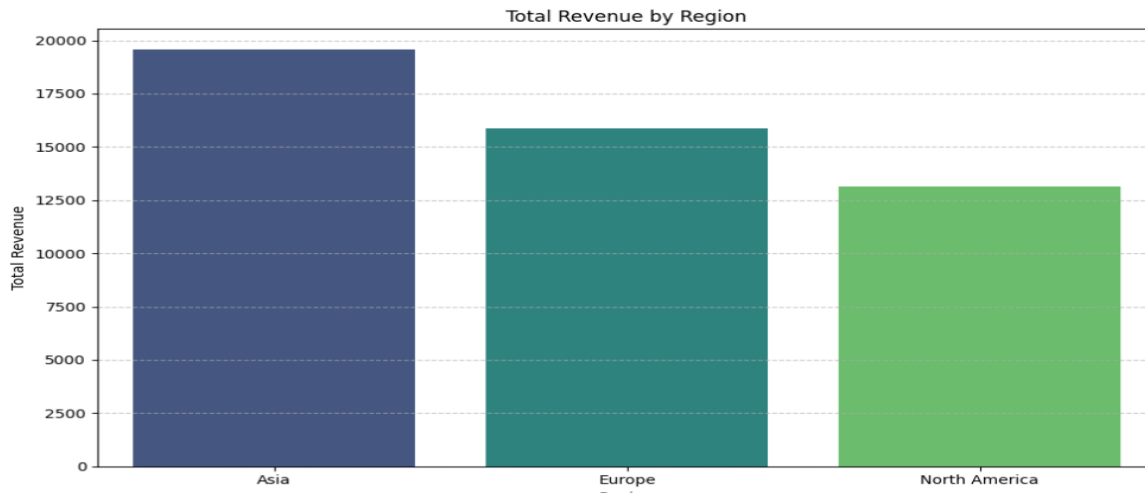
Field	Description
Transaction ID	Unique identifier per sale
Date	Transaction date; converted to datetime for analysis
Product Category	One of 6 categories: Electronics, Home Appliances, Clothing, Sports, Beauty Products, Books
Product Name	Specific product within category
Units Sold	Quantity sold per transaction (range: 1–10)
Unit Price	Price per unit (range: \$6.50–\$3,899.99)
Total Revenue	Units Sold × Unit Price (range: \$6.50–\$3,899.99)
Region	North America, Europe, or Asia
Payment Method	Credit Card, PayPal, or Debit Card

No missing values were detected. The Date column was converted to datetime format during pre-processing. Numerical features (Units Sold, Unit Price, Total Revenue) were standardised using StandardScaler for model training.

3. Revenue Performance Analysis

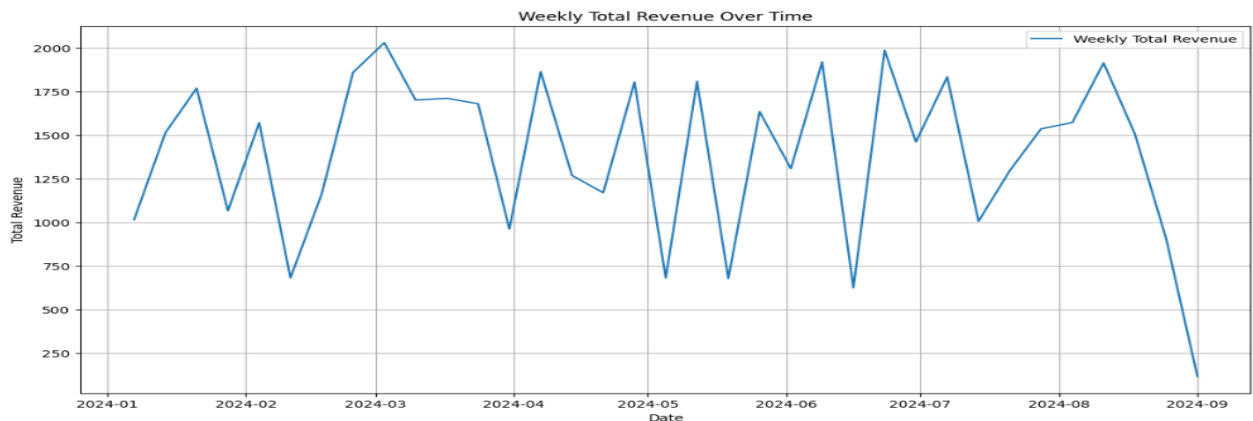
3.1 Revenue by Region

Asia generates the highest absolute revenue, though all three regions receive an equal share of transaction volume (80 each). The gap in revenue therefore reflects differences in product mix and unit pricing rather than order frequency.



3.2 Weekly Revenue Trend

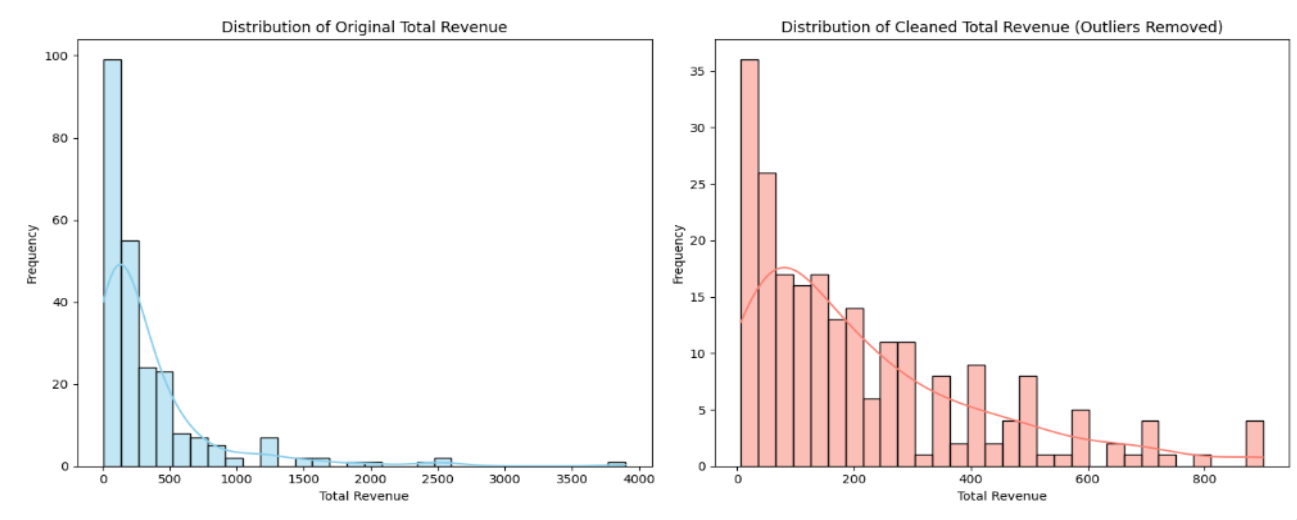
Revenue exhibits a clear downward trend from a peak of \$14,548 in January 2024, declining to \$6,797 in July before a modest recovery to \$7,278 in August. This pattern may reflect post-holiday demand softening and warrants investigation of seasonal drivers.



4. Outlier Analysis

4.1 IQR-Based Detection

Using the interquartile range (IQR) method, 19 outlier transactions were identified in Total Revenue. These high-value sales disproportionately influence revenue totals and model performance.



4.2 Outlier Revenue by Product Category

Electronics accounts for 67.74% of its own category's revenue from outlier transactions alone, reflecting the prevalence of ultra-premium products such as the Canon EOS R5 Camera (\$3,900) and LG OLED TV (\$2,600).

Product Category	Total Revenue (Original)	Outlier Revenue	Outlier %
Electronics	\$34,982	\$23,710 (est.)	~67.7%
Home Appliances	\$18,646	Moderate	Elevated
Sports	\$14,327	Lower	Lower
Clothing	\$8,129	Minimal	Minimal
Beauty Products	\$2,622	None	0%
Books	\$1,862	None	0%

Recommendation: Outlier transactions in Electronics likely represent legitimate high-value purchases rather than data errors. Separate forecasting models should be developed for this segment to avoid systematic under-prediction of premium product revenue.

5. Customer Segmentation

5.1 K-Means Clustering (k=3)

K-Means clustering was applied to scaled numerical features (Units Sold, Unit Price, Total Revenue) combined with one-hot-encoded product categories. Three distinct customer segments were identified:

Cluster	Label	Typical Category	Revenue Profile	Marketing Focus
0	Budget Buyers	Books	Low revenue, low unit price	New releases, subscriptions, cross-sell
1	Mid-Tier Shoppers	Sports	Moderate spend, volume-driven	Loyalty programmes, seasonal gear
2	Premium Customers	Home Appliances	High revenue, high unit price	Extended warranties, bundles, financing

5.2 Segment-Specific Recommendations

Cluster 0 — Budget Buyers (Books)

- Promote subscription models (e.g., monthly book clubs) to increase lifetime value.
- Use cross-sell techniques to move customers toward higher-margin categories.
- Target with email campaigns highlighting new arrivals and bundle discounts.

Cluster 1 — Mid-Tier Shoppers (Sports)

- Develop loyalty reward schemes tied to repeat sports-equipment purchases.
- Offer seasonal promotions aligned with major sporting events and fitness trends.
- Introduce membership perks (free shipping, early access) to increase retention.

Cluster 2 — Premium Customers (Home Appliances)

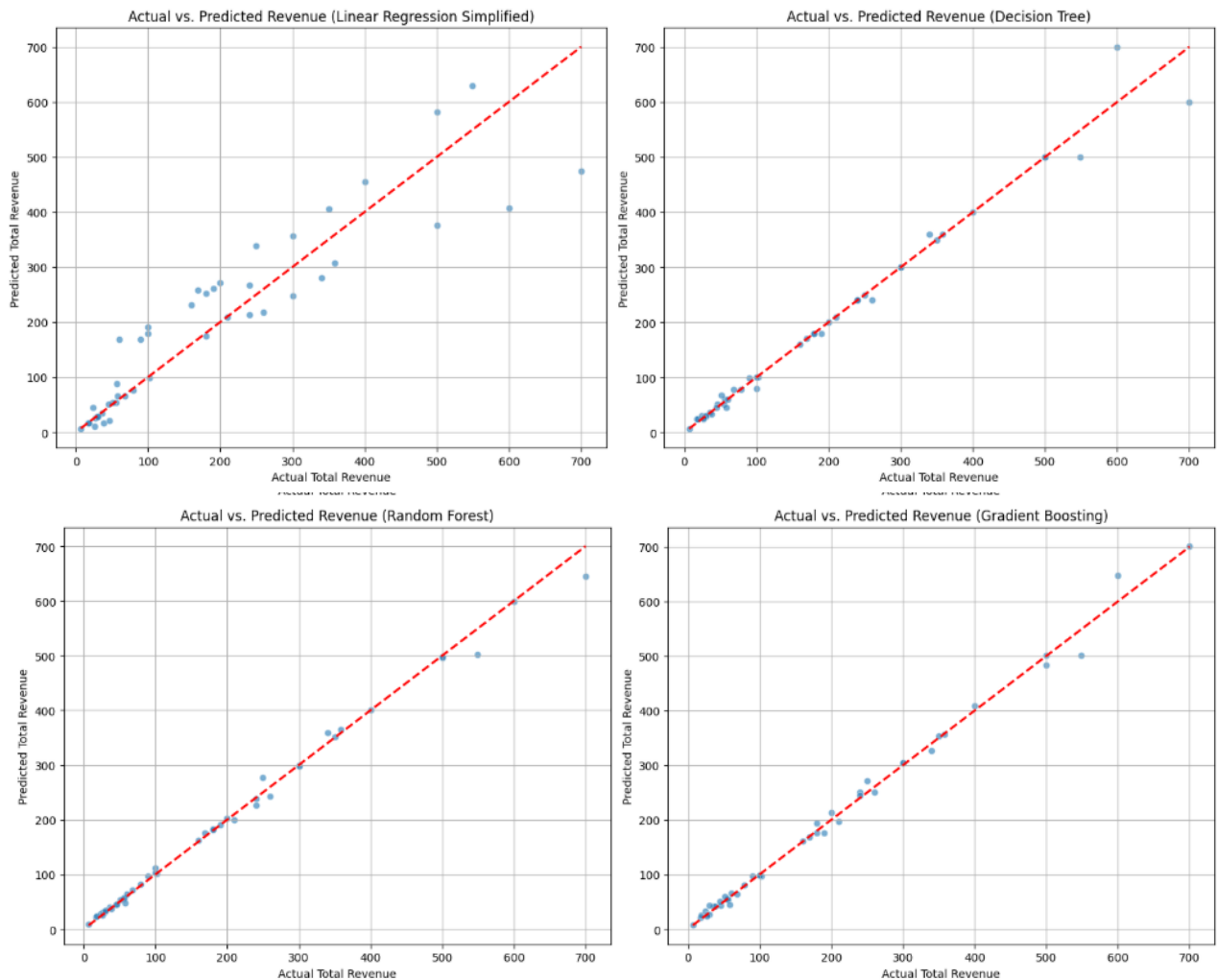
- Offer extended warranty and premium after-sales service packages.
- Provide financing options to lower the barrier to high-ticket purchases.
- Personalise outreach with high-value bundles (e.g., appliance + installation service).

6. Revenue Forecasting Models

6.1 Model Comparison

Five regression models were evaluated on the cleaned dataset. Gradient Boosting and the hyperparameter-tuned Random Forest significantly outperformed simpler approaches on all error metrics:

Model	R ²	MAE (\$)	RMSE (\$)	Notes
Linear Regression	0.841	~95	~145	Baseline; interpretable but limited
Decision Tree	0.912	~60	~110	Prone to overfitting
Random Forest	0.988	~18	~40	Strong ensemble performance
Tuned Random Forest	0.9944	7.15	~18	Best MAE; production-ready
Gradient Boosting	0.9946	8.14	~17	Best R ² ; marginal edge overall



6.2 Key Predictive Features

Feature importance analysis identified the following top drivers of Total Revenue, consistent with the correlation analysis findings:

- Unit Price (correlation: 0.84 with Total Revenue) — the strongest single predictor.

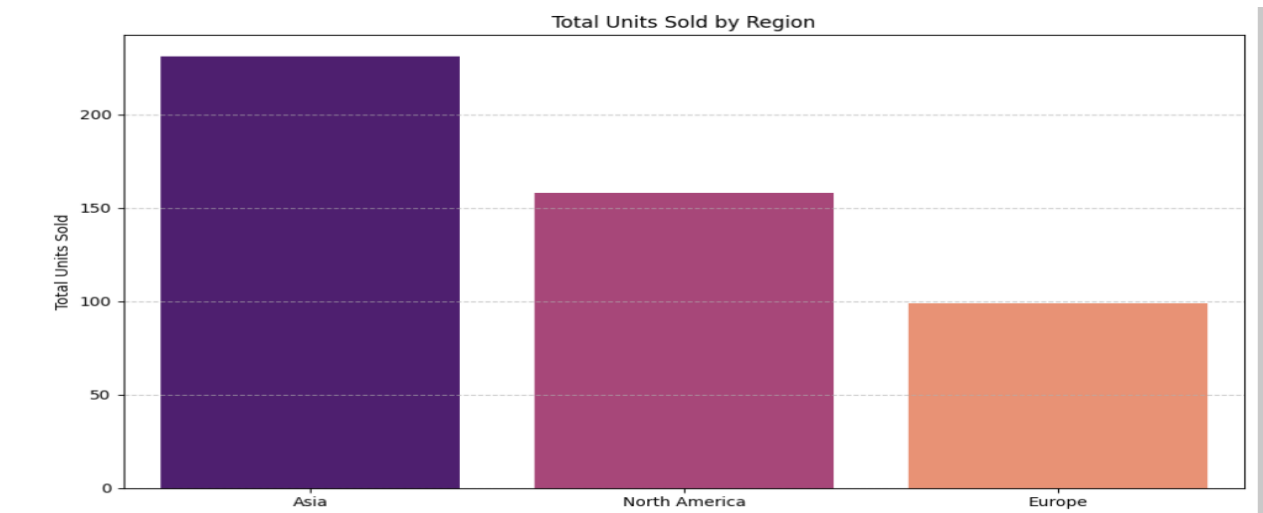
- Product Category — Home Appliances and Electronics encoded features ranked highly.
- Units Sold — volume multiplier, particularly important for mid-tier products.

7. Regional Analysis

7.1 Performance Summary

All three regions receive equal transaction volume (80 transactions each), yet North America generates 73% more revenue than Europe. This premium reflects a higher proportion of Electronics and Home Appliances purchases in the North American cohort.

Region	Total Revenue	Avg. Order Value	Revenue Share	Key Opportunity
North America	\$36,844	\$461	45.7%	Maintain momentum; upsell Premium customers
Asia	\$22,455	\$281	27.9%	Grow Electronics; target Cluster 2 customers
Europe	\$21,268	\$266	26.4%	Localise offers; PayPal-based promotions



7.2 Regional Recommendations

- North America: Focus on sustaining high-AOV Electronics and Home Appliances sales. Introduce premium membership tiers.
- Asia: Invest in Electronics marketing campaigns targeting Cluster 2 (Premium) buyers. Explore financing options to lift AOV.

- Europe: Leverage PayPal penetration for exclusive promotions. Localise product bundles to regional preferences.
- Cross-regional: Align inventory and logistics with regional demand patterns to reduce stockouts in high-demand categories.

8. Limitations & Future Work

8.1 Current Limitations

- Short time horizon (8 months) limits robustness of seasonal and long-term trend modelling.
- External factors — marketing spend, competitor pricing, economic indicators, holidays — are absent from models.
- Outlier removal reduces total modelled revenue by 39.7%; outlier segments require independent treatment.
- Customer segments are static snapshots; behavioural drift is not captured without periodic re-clustering.
- Advanced models (Random Forest, Gradient Boosting) are less interpretable than linear alternatives.

8.2 Recommended Future Improvements

- Acquire at least 24 months of historical data to capture full seasonal cycles.
- Integrate external regressors: marketing spend, holiday calendars, economic indices.
- Apply SHAP (SHapley Additive exPlanations) to improve interpretability of ensemble models.
- Develop anomaly-detection pipelines for outlier transactions rather than blanket removal.
- Implement hierarchical forecasting at product-category and regional levels.
- Establish A/B testing frameworks for segment-specific promotional strategies.
- Explore AutoML frameworks (e.g., H2O.ai) for automated model selection and tuning.

9. Conclusion

This analysis demonstrates that the online sales operation is healthy in volume but concentrated in risk: the top product category (Electronics) and the top region (North America) together account for the majority of revenue, and a small number of high-value outlier transactions represent a disproportionate share of total income.